

MVX — Developer Implementation Specification

Synthetic Dark Design System · Technical Hand-Off Document

M

MVX ENGINE
CORE_SPEC_BOARD

01

GLOBAL DESIGN TOKENS

COLOR PALETTE SYSTEM

Token	Role / Placement	Hex
☐ Midnight Navy	Foundation / Deep BG	#0A0E17
☐ Obsidian	Surface / Cards	#1F2937
☒ Electric Violet	Primary Accent / Glow	#8B5CF6
☒ Neon Cyan	Secondary Accent / Highlight	#06B6D4
☒ Near White	Primary Text Content	#F9FAFB
☐ Muted Silver	Secondary Prose / Detail	#94A3B8
☐ Toggle OFF	Disabled state track	#3C3C3C

TYPOGRAPHIC SCALE

Level	Weight	Size	Standard Usage
Display	Bold	32px	Page headers / MODULES
H1	Bold	24px	Sidebar branding titles
H2	Bold	18px	Individual module card titles
Body	Medium	13px	Sidebar categories
Caption	Regular	12px	Descriptions, footers, system tags
Micro	Medium	12px	Status indicators, values

BORDER RADII SPEC

Shell Window	20px
border-radius: 20px;	
Module Cards	12px
border-radius: 12px;	
Category Buttons	8px
border-radius: 8px;	
Precision Toggle	12px
border-radius: 12px;	

02

COMPONENT TECHNICAL SPECS

[COMP-01] GLASSMORPHIC SHELL

1280 x 800 px

Primary container framework utilizing internal overlays and backdrop filters.

- background: #0A0E17 solid
- border: 2px inside solid #8B5CF6
- corner-radius: 20px // overflow: hidden
- sidebar-bg: rgba(0, 0, 0, 0.3) | 260px fixed width

[COMP-02] MODULE CARD

300 x 100 px

Standard component blocks displaying system functionality with discrete controls.

- background: rgba(255, 255, 255, 0.05)
- border: 1px inside solid rgba(255, 255, 255, 0.10)
- padding: 20px left / 20px top
- origin-offsets: Title(20,20) | Desc(20,50) | Toggle(240,35)

[COMP-03] PRECISION TOGGLE

48 x 24 px

Surgical boolean control switches with smooth micro-interaction transitions.

- track: radius 12px | OFF State: #3C3C3C | ON State: #06B6D4
- knob: 18x18px circle | #FFFFFF | y-offset: 3px
- transition: 200ms ease-in-out

[COMP-04] CATEGORY BUTTON

220 x 44 px

Modular navigational items for quick sidebar category switches.

- font: Inter Medium 13px | #FFFFFF
- active-state: Fill #8B5CF6 @ 15% | Stroke 1px solid #8B5CF6
- inactive-state: No fill / transparent borders
- transitions: hover-fill #8B5CF6 @ 8% | 150ms ease-in-out

03

LAYOUT & GRID ARCHITECTURE

GRID SYSTEM BLUEPRINT

SHELL ANATOMY MAP

SCALE: 1:1 RESOLUTION

SIDEBARW: 260px
Pitch: 55px**COL 1**X: 300
W: 300px**COL 2**X: 620
W: 300px**COL 3**X: 940
W: 300px

SPATIAL ALIGNMENT SPECS

Sidebar & Navigation Pitch

Categories stacked vertically with exactly 11px spacing gaps. Total pitch of 55px (44px height + 11px padding gap).

Content Area Coordinates

Starts at absolute offset X: 300px. Standard header label 'MODULES' rendered at offset position (300, 40).

Module Card Grid Spacing

3-Column grid with column offsets at X: 300, 620, 940 and vertical rows at Y: 120, 240, 360. 20px gutter system.

04

INTERACTION & MOTION SPEC

TRANSITION SPECS

Trigger / Node	Duration	Easing Curve	Target CSS Properties
Toggle switch state	200ms	ease-in-out	background-color, transform
Category nav hover	150ms	ease-in-out	background-color, border-color
Category nav active	150ms	ease-in-out	background-color, border-color
Module card hover	200ms	ease-in-out	border-opacity, box-shadow
Shell window open/close	300ms	cubic-bezier	opacity, transform scale
Container content fade-in	250ms	ease-out	opacity

MICROY-INTERACTION REFINEMENTS

Module Card Hover State

Stroke opacity increases to 20%. Apply a subtle violet glow overlay.

box-shadow: 0px 0px 12px rgba(139, 92, 246, 0.20)

Active Toggle Scale Jump

Slight reactive scale snap on toggle hover.

transform: scale(1.05)

QUICK REFERENCE CHEAT SHEET

BG: #0A0E17 SURFACE: #1F2937 VIOLET: #8B5CF6 CYAN: #06B6D4 SHELL: 1280x800 r20 CARD: 300x100 r12 TOGGLE: 48x24 r12 GRID GAP: 20px